

Unit 4, Lesson 7: Applying Exponent Laws to Algebraic Expressions – Part 1

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Learning Target

- ☐ I can write an equivalent algebraic expression by applying the product of powers, power of a power, or power of a product laws of exponents.

4.7.1 Warm-Up: Patterns

1. The first three steps in a pattern are shown in the table. The pattern is created using yellow hexagons and blue squares.

		
Step 1	Step 2	Step 3

- a. Draw what you think Step 4 in the pattern will look like.
- b. Ibrahim said the pattern could be represented by the expression $1^n + 2^n$, where n is the number of the step in the pattern. Explain whether or not you agree with Ibrahim.

4.7.2 Guided Instruction: Equivalent Algebraic Expressions

Laws of exponents can be applied to generate equivalent algebraic expressions that are similar to numerical expressions.

1. A square is shown, where each side has a length represented by the expression $3a^5$.



The formula $A = s^2$ can be used to find the area of the square, where s is the side length of the square.

- a. Using this formula, the area of the square shown can be expressed as $(3a^5)^2$. Discuss with your partner how this expression represents the area of the square.
- b. Complete the steps to find an equivalent expression that represents the area of the square.

$$\begin{aligned}
 & (3a^5)^2 \\
 &= 3a^5 \cdot 3a^5 \\
 &= 3 \cdot 3 \cdot a^5 \cdot a^5 \\
 &= 3 \cdot 3 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot \boxed{} \\
 &= 9a \boxed{}
 \end{aligned}$$

- c. Suppose $a = 1.5$ units. Find the length of one side of the square using a calculator.**

- d. Use your answer to Part C to find the area of the square. Round to the nearest thousandth.**

- e. Use the expression generated in Part B to find the area of the square. Round to the nearest thousandth.**

- f. What do you notice?**

4.7.3 Exploration: Laws of Exponents with Algebraic Expressions

1. Several algebraic expressions are given in the table.
- a. For each expression, work with your partner to rewrite the expression in expanded form. Then, write an equivalent algebraic expression with one variable base and one coefficient. The first row has been partially completed.

Given Expression	Expanded Form	Equivalent Expression
$5c^2 \cdot c^6$	$5 \cdot c \cdot c \cdot c \cdot c \cdot c \cdot c \cdot c$	$5c \square$
$3x^5 \cdot 2z$		
$(2y^6)^2$		
$(m^2)^4$		
$(x \cdot y)^3$		
$(3c \cdot d)^2$		

- b. Discuss with your partner what you notice about how the coefficients and variable powers of the equivalent expressions compare to the given expressions.

- c. Gavin said properties of exponents could be applied to the original expressions to generate the same equivalent expression. Match each pair of expressions with the exponent law that applies. Then, complete the explanations.

Original Expressions	Which property of exponents applies?	Explanation
$5x^2 \cdot c^5$ and $3x^5 \cdot 2x$	☐ product of powers ☐ power of a product ☐ power of a power	The powers of the variable bases from the original expression were ☐ added to ☐ multiplied result in the power of the variable base in the equivalent expression.
$(2y^5)^2$ and $(m^2)^4$	☐ product of powers ☐ power of a product ☐ power of a power	The powers of the variable bases from the original expression were ☐ added to ☐ multiplied result in the power of the variable base in the equivalent expression.
$(x \cdot y)^2$ and $(3c \cdot d)^2$	☐ product of powers ☐ power of a product ☐ power of a power	Each factor in the expression within parentheses was raised to the ☐ power of the product. ☐ power of the power.

- d. Use words from the bank to complete the explanations of how the laws of exponents applied to the coefficients when generating the equivalent expressions.

multiplied	power	product	same
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In the expressions where product of powers law applied to the variable factors, the coefficients were _____ to find the coefficient of the _____.

In the expressions where the power of a power law applied, the coefficient was also raised to the power of the _____.

In the expressions where the power of a product law applied, the coefficients of the factors were also raised to the _____ power.

4.7.4 Bring It Together: Applying Laws of Exponents to Variables



The **product of powers**, **power of a power**, and **power of a product** laws of exponents apply to both variable and numerical bases when generating equivalent algebraic expressions.

1. Apply the law of exponents to generate equivalent expressions.

Product of Powers $a^m \cdot a^n = a^{m+n}$	Power of a Power $(a^m)^n = a^{mn}$	Power of a Product $(ab)^m = a^m \cdot b^m$
$x^7 \cdot x^9 =$ _____	$(g^2)^3 =$ _____	$(x^2y)^3 =$ _____
$2z^4 \cdot 5z^2 =$ _____	$(h^3)^6 =$ _____	$(4p^2)^3 =$ _____

4.7.5 Activity: Missing Numbers

1. Work with your partner to use the digits 0 – 9 to complete the equations. Each digit is used exactly once. **Blue boxes are coefficients** and **orange boxes are exponents**.

0	1	2	3	4	5	6	7	8	9
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a. $k^{\boxed{}} k^2 = k^7$

b. $(g^{\boxed{}})^{\boxed{}} = g^7$

c. $(\boxed{} x^4)^{\boxed{}} = 8x^{22}$

d. $r^{\boxed{}} \cdot r^5 \cdot r^3 = r^{\boxed{}}$

e. $(3c^{\boxed{}} \cdot b^3)^2 = \boxed{} c^3 \cdot b^{\boxed{}}$

4.7.6 Wrap-Up: Ticket Out the Door

1. Apply the laws of exponents to generate an equivalent algebraic expression for each of the following.

a. $(5y)^2 = \underline{\hspace{2cm}}$

b. $x^6 \cdot x^3 = \underline{\hspace{2cm}}$

c. $(t^5)^2 = \underline{\hspace{2cm}}$